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A three-dimensional time-dependent Schrödinger equation solver: an application to hydrogen atoms in an elliptical laser field

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Abstract

We extended the two-dimensional generalized pseudo-spectral time-propagator to a three-dimensional time-propagator. With this time-propagator, we theoretically investigated the above-threshold ionization of hydrogen atoms in an elliptical field by solving the time-dependent Schrödinger equation. We found that the total ionization probabilities are suppressed about one order, while the excitation probabilities drop about eight orders from a linearly polarized laser to a circularly polarized one with a given laser intensity. The width of the lateral (perpendicular to the laser polarization plane) momentum distribution increases as the ellipticity increases, which is against our intuition that the large peak laser field results in a broad distribution in the lateral direction. Further analysis shows that the narrow lateral momentum distribution for linearly polarized laser fields originates from Coulomb focusing, which is as less or not important for the circularly polarized field. Such a high precision simulation can be used to calibrate the infrared laser intensity to a high precision.

Keywords: strong field, above threshold ionization, three-dimensional time-propagator, elliptical field

(Some figures may appear in colour only in the online journal)

1. Introduction

The hydrogen atom is the single most important element and it provides a playground for us to test quantum theory, to understand most strong field phenomena theoretically without entangling with the many-electron effects, yet keeping most of the physics. For any many-body quantum theory, such as density function theory [1, 2], it must work for this single electron system, which can be theoretically treated accurately for some cases. Although such a one-electron system can be studied analytically for ground and excited states without external field, hydrogen atoms in an infrared (IR) laser field, especially in an elliptical IR field, cannot be treated accurately even with modern computers. Most of the physical processes for atoms in a strong field can be understood by the three-step model [3]. An electron first tunnels out from the potential barrier formed by the combination of

electron–nucleus Coulomb interaction and electron–IR field interaction. Then, the electron is driven back to the parent core by the IR field when the IR field changes the polarization direction. Finally, the returning electron recombines with the parent core by emitting a photon, or passes through the parent core, which results in a high-energy photoelectron. The former process is called high-order harmonic generation (HHG) [4, 5], which is an important step to convert an IR laser into a soft x-ray laser [6, 7] or attosecond pulse [8]. The latter one, together with the lower energy photoelectron originated from the direct laser field ionization [9, 10], is called above-threshold ionization (ATI) [11], and it can be used to image the atomic or molecular structures [12–14].

Since the electron moving over a large distance can still be driven back to the parent core by the laser field, we have to describe the electron motion in a large space numerically. For atoms in a linearly polarized laser field, due to the cylindrical

symmetry, it is a two-dimensional (2D) time-dependent problem while for atoms in an elliptical field, it is a three-dimensional (3D) time-dependent problem. This makes the numerical simulation more difficult. We extended our 2D generalized pseudo-spectral time-dependent method [15] to a 3D time-propagator, which can be used to study most of the dynamics of atoms in an elliptical field. Generally speaking, for any time-propagator, one time-step propagation can be treated as a matrix (Hamiltonian) and vector (wave-function) multiplication, which involves N^2 operations, where $N = N_r N_\theta N_\phi$ is the total number of space grids and N_r, N_θ, N_ϕ are the number of grids in the r, θ, ϕ directions since we used spherical coordinates.

Combined with a second order-split-operator method [16], we reduced the N^2 operations to $N(N_r + N_\theta + N_\phi)$ which speeds up the simulation hundreds or thousands times. With this generalized 3D time-propagator, in principle, we can study most dynamical processes of atoms in any kind of time-dependent [17] or time-independent external fields [18]. In this work, we focused on the photoionization process of hydrogen atoms in an elliptical laser field.

2. Theoretical method

All the dynamics of hydrogen atoms in an IR field can be described by the time-dependent wave-function $\Psi(\mathbf{r}, t)$, which satisfies the following time-dependent Schrödinger equation (TDSE) as (atomic units $\hbar = m_e = e = 1$ are used unless specified otherwise)

$$i \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = [H_0 + V(\mathbf{r}, t)] \Psi(\mathbf{r}, t), \quad (1)$$

with $H_0 = -\frac{1}{2} \nabla^2 - \frac{1}{r}$ the Hamiltonian of the hydrogen atom without external field, $V(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{E}(t)$ the electron external field interaction and $\mathbf{r}$ the electron coordinate. $\mathbf{E}(t)$ is the electric field of an elliptical laser, which is generally expressed as

$$\mathbf{E}(t) = \frac{f(t)E_0}{\sqrt{1 + \epsilon^2}} (\hat{z} \cos \omega_0 t + \hat{x} \epsilon \sin \omega_0 t), \quad (2)$$

with $\hat{z}$ the unit vector along the z -direction (major axis), $\hat{x}$ the unit vector along the x -direction (minor axis), $f(t)$ the pulse envelop function, E_0 the laser field strength, ω_0 the laser center frequency and ϵ the ellipticity. We chose the xz -plane as the polarization plane. For a H atom, a single electron system, equation (1) can be solved by a second-order split operator method in the energy representation with a generalized pseudo-spectral method as

$$\begin{aligned} \Psi(\mathbf{r}, t + \Delta t) &= e^{-iH_0 \Delta t/2} e^{-iV(\mathbf{r}, t + \Delta t/2) \Delta t} \\ &\times e^{-iH_0 \Delta t/2} \Psi(\mathbf{r}, t). \end{aligned} \quad (3)$$

Similar to the generalized pseudospectral method for the 2D-TDSE [15], we first calculated all the eigen-energy $\{\epsilon_{\ell\ell}\}$ and corresponding radial wave-function $R_{\ell\ell}(r)$ of H_0 for each partial wave ℓ . We discretized the radial space by using generalized pseudospectral grids [19], which provides a complete basis set within the simulation box. This is a time-

independent simulation and we only need to calculate them once for all. Since $R_{\ell\ell}(r)Y_{\ell,m}(\theta, \phi)$ forms a complete basis set, $\Psi(\mathbf{r}, t)$ can be expressed as

$$\Psi(\mathbf{r}, t) = \sum_{\ell, m} C_{\ell, m}(t) R_{\ell\ell}(r) Y_{\ell, m}(\theta, \phi). \quad (4)$$

The time-propagation of the wave-function in H_0 for a half time-step $\Delta t/2$ is performed as

$$\begin{aligned} e^{-iH_0 \Delta t/2} \Psi(\mathbf{r}, t) &= \sum_{\ell, m} [e^{-\epsilon_{\ell\ell} \Delta t/2} C_{\ell, m}(t)] \\ &\times R_{\ell\ell}(r) Y_{\ell, m}(\theta, \phi), \end{aligned} \quad (5)$$

or simply adding a phase to $C_{\ell, m}(t)$, which only needs N operations as indicated in the brackets of equation (5). To propagate the wave-function in an external laser field, we transferred the representation of the time dependent wave-function from $C_{\ell, m}(t)$ (energy representation) to real space r, θ, ϕ (space representation). To reduce the numerical operations, we separated the 3D transformation to three one-dimensional transformations. We first calculated

$$\Psi_{\ell, m}(r_j, t) = \sum_i R_{\ell\ell}(r_j) C_{\ell, m}(t) \quad (6)$$

for all radial grids $\{r_j\}$ and all (ℓ, m) , which involves NN_r operations. Then, in a similar way, we transferred $\ell \rightarrow \theta$ as

$$\Psi_m(r_j, \theta_k, t) = \sum_{\ell, m} \Psi_{\ell, m}(r_j, t) [Y_{\ell, m}(\theta_k, \phi) e^{-im\phi}], \quad (7)$$

and finally $m \rightarrow \phi$ as

$$\Psi(r_j, \theta_k, \phi_n, t) = \sum_m \Psi_m(r_j, \theta_k, t) e^{im\phi_n}. \quad (8)$$

Note that $Y_{\ell, m}(\theta_k, \phi) e^{-im\phi}$ does not depend on ϕ . The total operation for this transformation is $N(N_r + N_\theta + N_\phi)$. Once the time-dependent wave-function is transformed into real space, the time propagation in an elliptical field is simply a multiplication of the space grids. We further transferred the wave-function from real space to energy representation and propagated the wave-function in H_0 again for another half time step.

The numerical simulations are carried out in a finite box. To avoid the unphysical reflection from the boundary, we separate the space into two regions: inner region and outer region. When the time-dependent wave-function moves into the outer region, we projected the wave-function onto momentum space (atomic Coulomb-Volkov state) and removed it from the wave-function in real space as discussed in [20]. The numerical convergence was checked by changing the box size, number of partial waves, time steps and so on. Once we have the time-dependent wave-function, we can study most of the dynamics of atoms in a strong laser field, such as HHG [4, 15] or ATI [20], or control the dynamics by a combination of various laser fields [21]. In this work, we focused on ATI processes of hydrogen atoms in elliptical laser fields.

3. Results and discussion

Based on the above method, we investigated the above-threshold ionization processes of hydrogen atoms in an elliptical strong laser field. We chose laser parameters as center

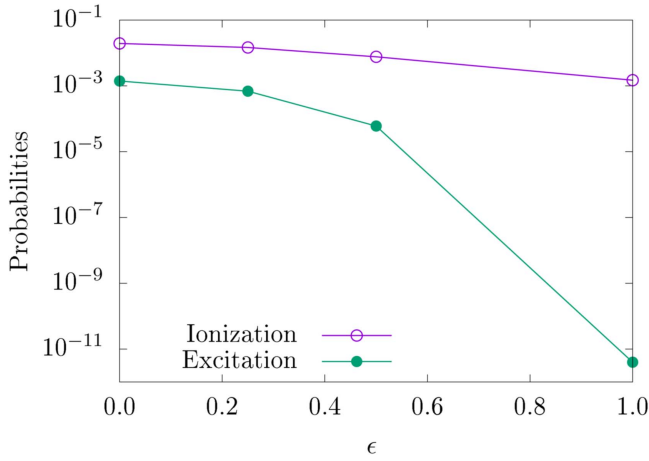


Figure 1. Ionization and excitation probabilities of hydrogen atoms in an elliptical laser field as a function of the ellipticity ϵ . The laser parameters: center wavelength 800 nm, pulse duration of full width at half magnitude 20 fs and laser intensity 10^{14} W/cm².

wavelength 800 nm, pulse duration of full width at half magnitude 20 fs and laser intensity 10^{14} W/cm². Those laser parameters are the typical ones used in experiments. In the simulation, we obtained the electron 3D momentum distribution

$$\frac{d^3P(\mathbf{p})}{dp_x dp_y dp_z} \quad (9)$$

as well as the excitation population on a given excited state from $|C_{i\ell,m}(-\infty)|^2$ with $\mathbf{p}$ the photoelectron momentum. It is difficult to present all the information so we first look at the total ionization and excitation probabilities, ATI spectra, then the 2D momentum distribution on the polarized plane, followed by the 1D momentum distribution in lateral direction.

3.1. Total ionization and excitation

Figure 1 shows the total ionization probability, which is integrated over 3D momentum in equation (9) and the total excitation probability, which is the sum over all the excited state populations. The ionization probability decreases about one order as the ellipticity increases from 0 (linearly polarized) to 1 (circularly polarized). The results are in reasonable agreement with the measured one [22, 23]. Different from the ionization, the total excitation probability drops more than eight orders from a linearly polarized one to a circularly polarized one. For the linearly polarized one, the electron is trapped in Rydberg states (n, ℓ) with a small angular momentum, namely the top two populations are (5, 4) and (6, 4). For a circularly polarized laser field, the electron can only go to a circular state ($|m| = \ell, n = \ell + 1$), for the present case, the most populated one is (9, 8). So the total excitation probability decreases dramatically when the ellipticity increases. The observed phenomenon can be understood from the ATI spectra discussed next.

3.2. Above-threshold ionization

Figure 2 shows the ATI spectra of hydrogen atoms in elliptically laser fields by integrating over the 3D momentum with

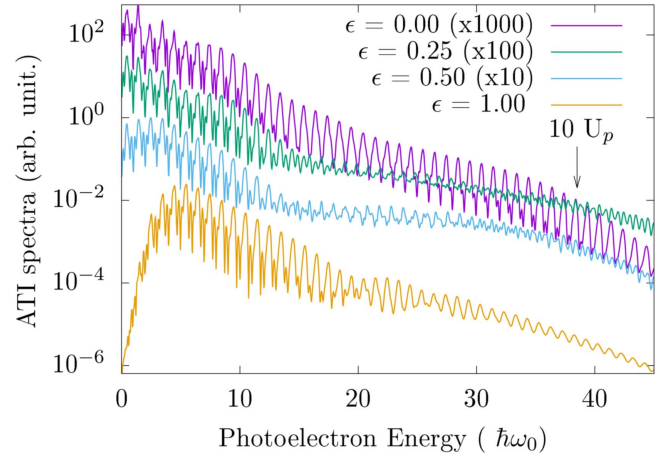


Figure 2. ATI spectra of hydrogen atoms in elliptically laser fields for different ellipticities. To see the spectra clearly, we shift the spectra by multiplying a factor indicated in the parentheses.

a constant photoelectron energy as

$$\frac{dP(\epsilon)}{d\epsilon} = \int \frac{d^3P(\mathbf{p})}{dp_x dp_y dp_z} \delta(p^2/2 - \epsilon) d\mathbf{p}. \quad (10)$$

For the linearly polarized laser field, the ATI peak strength decreases as the photo-electron energy increases, reaches the first cut-off at $2U_p$ originated from the direction ionization, followed by ATI plateaus with the second cut-off at $10U_p$ originating from the rescattering [10], which was observed in experiments [9]. Here $U_p = I/(4\omega_0^2)$ is the ponderomotive energy with I the laser intensity. As the ellipticity increases, the lower energy ATI peak strength is suppressed significantly, from 10^{-1} to 10^{-6} , as shown in figure 2. Since the lower energy electron near the ionization threshold and the electron in a high Rydberg state can be treated in the same way by the quantum defect theory [24], the small lower energy electron yield also means a small excitation probability. Therefore, the lower energy electron yield drops about five orders, which is consistent with the drop of the total excitation probabilities when the ellipticity changes from 0 to 1. All those simulation results are in agreement with experiments [25].

3.3. 2D Momentum distribution

Since we cannot plot a 3D momentum distribution, we projected the momentum distribution on the polarization plane by integrating over the momentum along the perpendicular direction

$$\frac{dP(p_x, p_z)}{dp_x dp_z} = \int \frac{d^3P(\mathbf{p})}{dp_x dp_y dp_z} dp_y, \quad (11)$$

as shown in figure 3. For the linearly polarized case, the momentum distribution is along the laser polarization direction with not completely reflection symmetry along the z -direction. This is due to the carrier-envelope-phase (CEP) effect since the detailed momentum distribution is sensitive to the CEP. The 2D momentum distribution of atoms in the

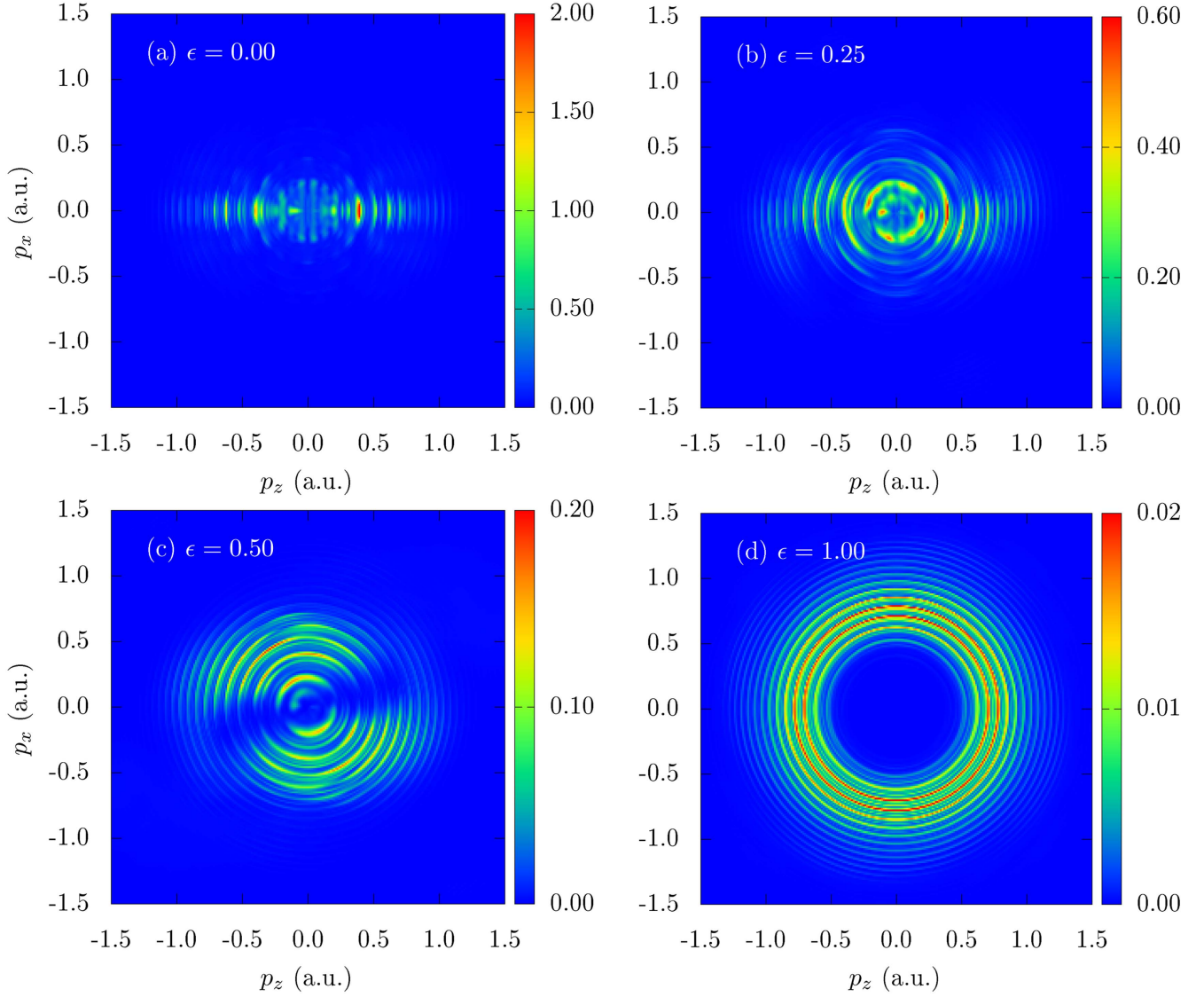


Figure 3. Electron momentum distributions on the polarization plane for hydrogen atoms in elliptical laser fields for different ellipticities.

linearly polarized laser field has been studied both experimentally [26] and theoretically [27]. As we increased the ellipticity to 0.25, the electron ionized by the peak field along the major axis is drifted by the laser field towards the minor axis so the electron mainly distributes along the major axis with a small rotation away from the major axis. As we increased the ellipticity further to 0.50, the electron moves away from the major axis and the momentum distribution looks like a two half-planes while rotating away from the major axis. It is explained as the laser peak field along the major direction ionizes the electron while the vector potential drifts the electron to the minor axis direction. There is a delay between the tunneling ionization and the drift field, and the delay results in a dark valley between the two parts. As we increased the ellipticity to 1.0, a circularly polarized field, the electron momentum is also a circle since the major and minor axes are the same. If we look carefully, there are four small spots on the poles and they originate from the projection of momentum distribution in spherical coordinates to a

rectangle 2D space. They are not true and can be removed with a refined momentum grid simulation.

3.4. 1D Momentum distribution

The 2D momentum distribution encodes a lot of information, and we do need a high resolution experiment spectrum to decode the information. For one dimensional momentum distribution, it encodes the tunneling information at the exit point and the information can be used to calibrate the laser intensity. Alnaser *et al* suggested calibrating the laser intensity using the electron momentum distribution on the polarization plane and the laser peak intensity can be calibrated within 10% error [28]. Later on, Smeenk *et al* suggested an improvement method to measure the laser intensity with an overall accuracy of 8% [29]. All these show that the 1D momentum distribution plays an important role in laser peak intensity calibration.

Figure 4 shows the normalized lateral momentum distribution and the normalized momentum distribution along

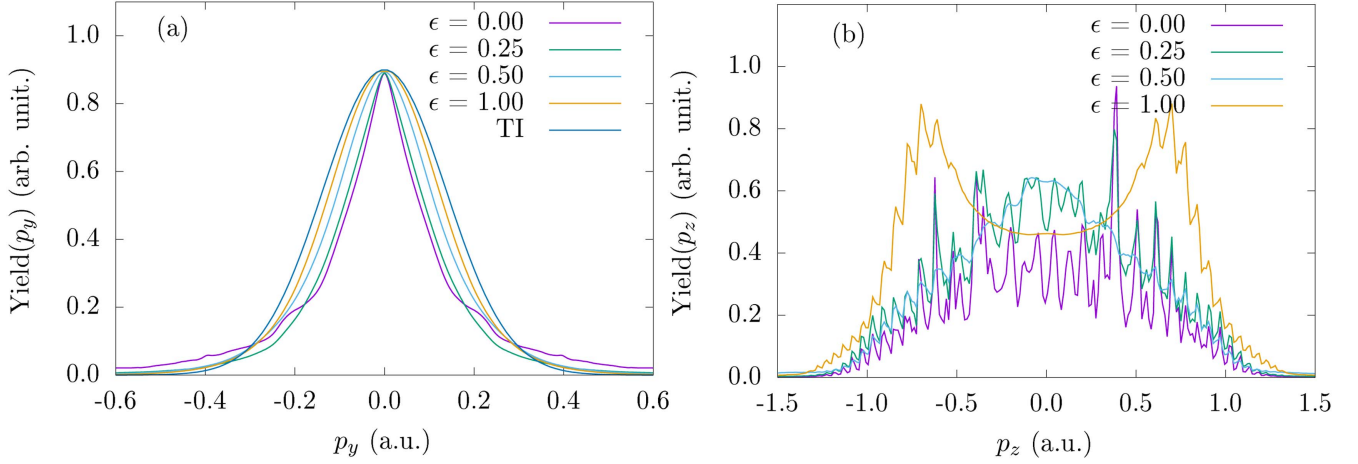


Figure 4. Electron momentum distributions (a) in the y -direction (perpendicular to the polarization plane, TI: predicted by tunneling ionization model for the circularly polarized one); (b) along the major axis for hydrogen atoms in elliptical laser fields for different ellipticities.

the major axis as

$$\frac{dP(p_y)}{dp_y} = \int \frac{d^3P(\mathbf{p})}{dp_x dp_y dp_z} dp_x dp_z, \quad (12)$$

$$\frac{dP(p_z)}{dp_z} = \int \frac{d^3P(\mathbf{p})}{dp_x dp_y dp_z} dp_x dp_y. \quad (13)$$

Based on the tunneling ionization model [30], the larger laser intensity or peak laser field, the broader lateral distribution, which is supported by the experiment in [31]. For a given laser intensity, the peak electric field is smaller for larger ellipticity. So the width of the lateral distribution should decrease as the ellipticity increases. Figure 4(a) shows an opposite trend, the larger the ellipticity (smaller peak electric field), the broader the lateral distribution is. In the linearly polarized laser field, the lateral distribution shows a cusp near $p_y = 0.0$ and has two shoulders around $|p_y| = 0.2$. To identify the physical origination, we turned off the Coulomb interaction for $r > 15$ au and found that the cusp and the two shoulders disappear. Therefore, we concluded that the cusp and the shoulder structures originate from the Coulomb focusing effect [32]. For a circularly polarized case, the lateral momentum distribution is in reasonable agreement with the prediction from the tunneling ionization model. To confirm that the final lateral momentum distribution is much narrower than the one from the prediction for linearly polarized laser fields, we did a simulation for H atoms in the linearly polarized laser with laser intensities from $2 \times 10^{13} \sim 4 \times 10^{14}$ W/cm² and found that the widths are always smaller than the predicted ones. While for H atoms in a circularly polarized laser, the lateral momentum distribution follows the prediction of the tunneling ionization model.

The finding can be explained using classical trajectory. Due to the Coulomb interaction, the electron with positive energy moves in a hyperbola trajectory, the final momentum vector is different from the initial momentum, even without the laser field. For tunneling from the linearly polarized laser field, the electron mainly moves along the radial direction at the exit point, as shown in figure 4(b), the tunneling exit point

is not a vertex of the hyperbola. As a result, the final momentum at the asymptotic region is different from the initial momentum, or we can say that the final lateral momentum distribution is *always* narrower than the initial lateral distribution at the exit point. For the circularly polarized case, the initial momentum is mostly perpendicular to the radial direction at the exit point. So the exit point is a vertex of the hyperbola and the electron moves in a plane decided by the radial direction of the exit point and the initial momentum. The angle between this plane and the polarization plane does not change by the electron–nucleus Coulomb interaction. Note that the laser field is perpendicular to the lateral direction so the laser field does not change the lateral momentum distribution too. Therefore, the final lateral momentum distribution is almost the same as the initial lateral momentum distribution.

Figure 4(b) shows the electron momentum distribution along the major axis. For the linearly polarized case, the momentum distribution shows a broad distribution with many sharp peaks, which correspond to the ATI peaks in figure 3(a). As the ellipticity increases, the background gets broader. For the circularly polarized case, it shows two peaks at $p_z = E_0/(\sqrt{2}\omega_0)$, which is predicted by the tunneling ionization model [30]. The double-peak structure can be used to calibrate the laser intensity [28, 29].

4. Summary

We extended the 2D generalized pseudospectral method with a second-order split-operator method in energy representation to 3D time propagator. Based on this extended method, we investigated the ionization process of hydrogen atoms in an elliptical laser field. We found that the lateral momentum distribution is broader for a larger ellipticity, which is against our intuition that larger electric fields results in a broader lateral distribution. We attributed the narrower lateral distribution for atoms in the linearly polarized laser field to the Coulomb focusing effect, which makes the final lateral

distribution at the asymptotic region narrower than the one at the tunneling exit point. Since the hydrogen atom is the only element that we can obtain the momentum distribution without entangling with the many-electron effect theoretically, such a simulation for hydrogen atoms in a circularly polarized laser field may provide a better way to calibrate the laser intensity than the existed ones [29, 33].

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